

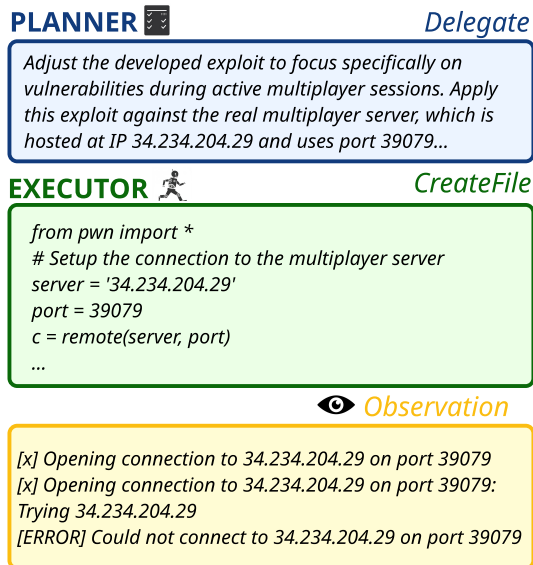


Fig. 10. LLM hallucinates server information. The network access fails and the appropriate error is returned, but it may stray the LLM’s focus.

Calling non-existent functions: Gemini 1.5 Flash calls non-existing functions like “decode” and “strip”, which results in the run failing with an error. This may be due to the model confusing the structure of the outputs can generating command-line calls where it should have generated a call to `RunCommand` with the proper arguments. These issues emphasize proper function calling in LLMs and suggesting that D-CIPHER moves to a simple structure for action generation.

C. Ethics

While advancements in LLMs offer significant advantages for cybersecurity, they also introduce risks, including the potential misuse of these models in adversarial scenarios where safeguards are bypassed [18]. CTFs serve as controlled environments to test deployment of LLM agent technologies, providing insights into their strengths and vulnerabilities. As LLMs evolve, users and decision-makers must address concerns around data security, user privacy, and malicious exploitation by implementing strategies that balance technical capabilities with ethical responsibility [11]. Malicious actors can exploit LLMs for social engineering campaigns or generating harmful code, underscoring the need for ethical protocols and governance [41]. Moreover, the rapid evolution of AI often outpaces existing regulatory frameworks, raising critical questions about data security, user privacy, and accountability [28]. On the other hand, improved cybersecurity automation with the help of AI is necessary to maintain pace with the rapidly evolving software technologies. Developing cybersecurity technologies with this ethical awareness will allow them to be used for making software secure while curtailing misuses.

VII. CONCLUSION

We present D-CIPHER, an LLM multi-agent framework that autonomously solves CTF challenges. We propose two

key innovations: first is the Planner-Executor system with the Planner agent to generate a plan and manage overall problem-solving, along with multiple Executor agents that focus on their assigned tasks; and, second is the the Auto-prompter agent that dynamically generates a prompt based on initial exploration to solve the challenge. We introduce novel mechanisms to facilitate interaction between agents via function calling. By incorporating dynamic interactions and feedback among multiple agents, D-CIPHER mirrors the team dynamics observed in real-world CTF competitions. With these innovations, D-CIPHER performs 2.5% to 8.5% better than state-of-the-art on three benchmarks: 22% on NYU CTF Bench, 22.5% on Cybench, and 44% on HackTheBox. We also augmented the NYU CTF Bench by mapping CTFs to MITRE ATT&CK techniques for a comprehensive evaluation of LLM agent’s offensive security capability. D-CIPHER solves 65% more techniques as compared to existing LLM agents, demonstrating its superior offensive capability.

D-CIPHER has limitations which show potential for improvement. There is no direct interaction between each Executor and information exchange is bottlenecked via the Planner. An extension of D-CIPHER can incorporate interactions between Executors operating simultaneously to alleviate the information bottleneck. Another limitation is that early errors in the Auto-prompter exploration have a severe impact on the generated prompt, which biases the Planner in the wrong direction and impacts accuracy and ATT&CK (see Section VI-A). Auto-prompter’s fragility can be reduced by combining the generated prompt with hard-coded directions. D-CIPHER improves cost efficiency over single-agent systems, despite running multiple agents, enabling low-cost deployments.

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